

Module 15 - chapter 15

Ansible Configuration - Default Inventory File



NOTE:

This is the last time we use Nexus in the Bootcamp, which means you can power down and destroy the server in order to avoid any more additional costs.

Configure Git

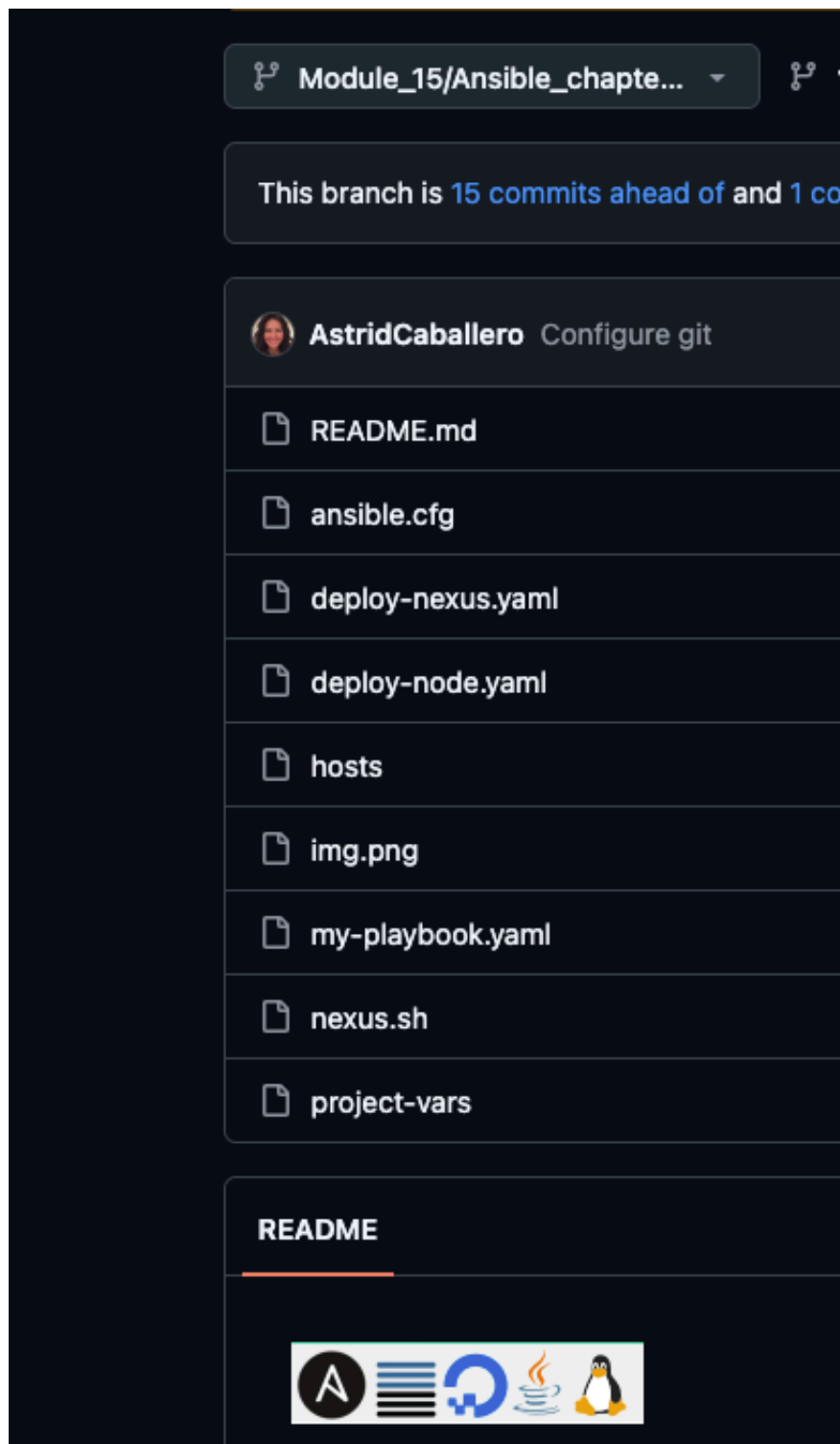
The screenshot shows a VS Code editor on the left with a file explorer for an 'ANSIBLE' project. The files listed are: `ansible.cfg`, `deploy-nexus.yaml`, `deploy-node.yaml`, `hosts`, `my-playbook.yaml`, `nexus.sh`, and `project-vars`. On the right, a 'Setup Git Project' overlay is displayed. It features a red Git logo at the top right. The main content area lists four bullet points: 'Create a Git Repository', 'Your Playbooks should be stored safely in this remote repository', 'Git repo is your "single source of truth" for your Ansible playbooks', and 'Great for working in teams'. To the right of these points is a GitLab logo and a document icon with a circled 'A'. Below the list, there is a red button that says 'Check out Git Module for setup demo'. At the bottom right, there are two icons: a person icon and a server rack icon, both with a circled 'A'.

Setup Git Project

- ▶ Create a Git Repository
- ▶ Your Playbooks should be stored safely in this remote repository
- ▶ Git repo is your "single source of truth" for your Ansible playbooks
- ▶ Great for working in teams

Check out Git Module for setup demo

done:



Configure Inventory Default location

We now will configure the default host file, so far when we execute the playbook we have to pass the 'hosts' file as a parameter in the terminal:

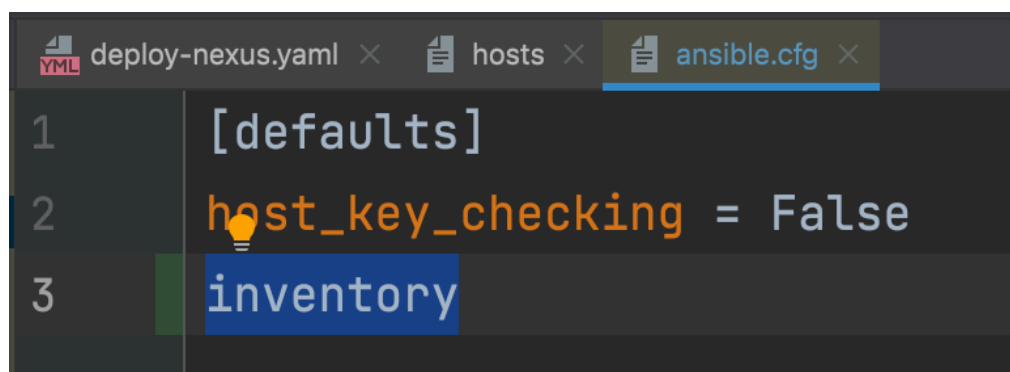
```
ansible-playbook -i hosts deploy-nexus.yaml
```

If we make the 'hosts' file default then we don't have to pass it as a parameter and we can just execute the playbook with only the playbook name

-> ansible-playbook <playbook_name>

-> ansible-playbook deploy-nexus.yaml

For that we edit the 'ansible.cfg' file where we now will configure a default 'hosts' file location using the official term 'inventory'

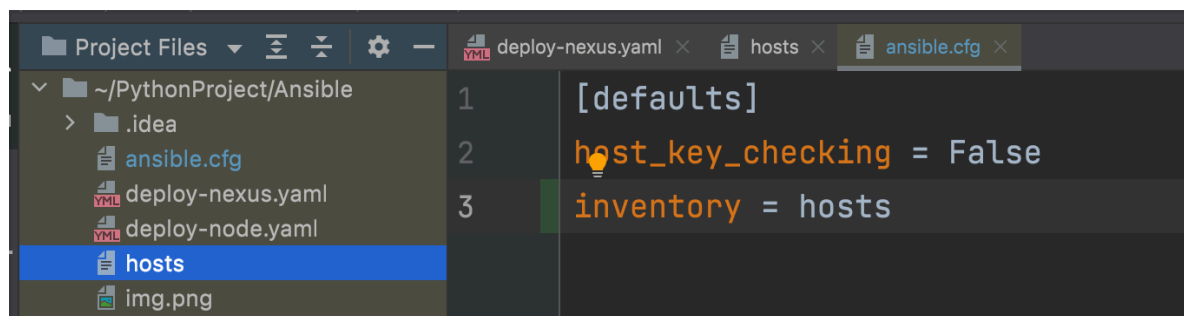


The screenshot shows an IDE with three tabs: 'deploy-nexus.yaml', 'hosts', and 'ansible.cfg'. The 'ansible.cfg' tab is active, showing the following content:

```
1 [defaults]
2 host_key_checking = False
3 inventory
```

That represents the server or list of servers configured in the 'hosts' file and then we pass the location of the 'hosts' file.

And because 'ansible.cfg' and 'hosts' files are in the same level (same folder) we just type the name of the hosts file which is 'hosts'



The screenshot shows an IDE with three tabs: 'deploy-nexus.yaml', 'hosts', and 'ansible.cfg'. The 'ansible.cfg' tab is active, showing the following content:

```
1 [defaults]
2 host_key_checking = False
3 inventory = hosts
```

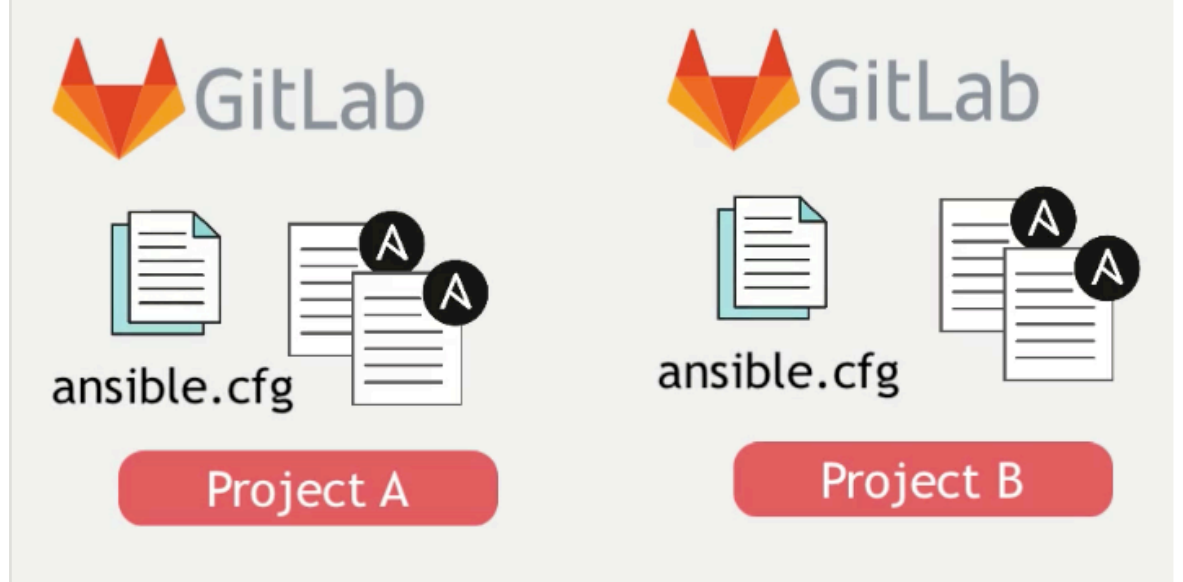
The file explorer on the left shows the following files:

- ~/PythonProject/Ansible
 - .idea
 - ansible.cfg
 - deploy-nexus.yaml
 - deploy-node.yaml
 - hosts
 - img.png

Now ansible will look for the specify 'hosts' file to get the inventory of servers.

This is helpful to have different configuration for each project

- Have an inventory file for each project



Let's test the playbook

-> ansible-playbook deploy-nexus.yaml

```
> ansible-playbook deploy-nexus.yaml

PLAY [Install Java and net-tools] *****

TASK [Gathering Facts] *****
ok: [188.166.148.119]

TASK [Update apt repo and cache] *****
ok: [188.166.148.119]
```

If 'gathering facts' is OK that means that ansible was able to find the right 'hosts' file

```
TASK [debug] *****
ok: [188.166.148.119] => {
  "msg": [
    "nexus      8922  143  6.2 5447092 252132 ?      Sl   10:29   0:05 /opt/nexus/jdk/temurin_17.0.13_11_1
17.0.13+11/bin/java -server -Dnexus.installer.type=linux-x86-64 -Xms2703m -Xmx2703m -XX:+UnlockDiagnosticVMOpti
tput -XX:LogFile=./sonatype-work/nexus3/log/jvm.log -XX:-OmitStackTraceInFastThrow -Dkaraf.home=. -Dkaraf.base
ogging.config.file=etc/spring/java.util.logging.properties -Dkaraf.data=./sonatype-work/nexus3 -Dkaraf.log=./
xus3/log -Djava.io.tmpdir=./sonatype-work/nexus3/tmp -Djdk.tls.ephemeralDHKeySize=2048 -Dfile.encoding=UTF-8 -
xml=java.logging --add-opens java.base/java.security=ALL-UNNAMED --add-opens java.base/java.net=ALL-UNNAMED --a
se/java.lang=ALL-UNNAMED --add-opens java.base/java.util=ALL-UNNAMED --add-opens java.naming/javax.naming.spi=A
-opens java.rmi/sun.rmi.transport.tcp=ALL-UNNAMED --add-exports=java.base/sun.net.www.protocol.http=ALL-UNNAMED
ava.base/sun.net.www.protocol.https=ALL-UNNAMED --add-exports=java.base/sun.net.www.protocol.jar=ALL-UNNAMED --
xml.dom/org.w3c.dom.html=ALL-UNNAMED --add-exports=jdk.naming.rmi/com.sun.jndi.url.rmi=ALL-UNNAMED --add-export
sasl/com.sun.security.sasl=ALL-UNNAMED --add-exports=java.base/sun.security.x509=ALL-UNNAMED --add-exports=java
ty.rsa=ALL-UNNAMED --add-exports=java.base/sun.security.pkcs=ALL-UNNAMED -jar /opt/nexus/bin/sonatype-nexus-rep
1.jar",
    "root      9055  0.0  0.0   2800   1912 pts/0    S+   10:29   0:00 /bin/sh -c ps aux | grep nexus",
    "root      9057  0.0  0.0   7076   2164 pts/0    S+   10:29   0:00 grep nexus"
  ]
}
```

```
TASK [Wait one minute] *****
Pausing for 60 seconds
(ctrl+C then 'C' = continue early, ctrl+C then 'A' = abort)
ok: [188.166.148.119]

TASK [Check with netstat] *****
changed: [188.166.148.119]

TASK [debug] *****
ok: [188.166.148.119] => {
  "msg": [
    "Active Internet connections (only servers)",
    "Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name    ",
    "tcp        0      0 0.0.0.0:54:53           0.0.0.0:*               LISTEN      712/systemd-resolve ",
    "tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN      1/init              ",
    "tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN      712/systemd-resolve ",
    "tcp6       0      0 :::8081                 :::*                    LISTEN      8922/java            ",
    "tcp6       0      0 :::22                   :::*                    LISTEN      1/init              "
  ]
}
```

```
PLAY RECAP *****
188.166.148.119      : ok=24   changed=13   unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

And we check the droplet

```
root@nexus:~# netstat -pnlt
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 127.0.0.54:53           0.0.0.0:*               LISTEN      712/systemd-resolve
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN      1/init
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN      712/systemd-resolve
tcp6       0      0 :::8081                 :::*                    LISTEN      8922/java
tcp6       0      0 :::22                   :::*                    LISTEN      1/init
root@nexus:~#
```

And everything is working 🥳